STATISTICS

Chapter 1 Homework

Sampling, data, and study design

Sections 1.1-1.4 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Populations, samples, and variables

1. A college randomly selects 80 of its 2,000 students to ask whether they used the tutoring center this quarter. Of those selected, 28 say yes. Identify the population, sample, variable, and observed data type. Compute the sample proportion and distinguish it from the unknown population proportion.

2. In Problem 1, distinguish (a) reporting that 35% of the sample used tutoring from (b) estimating tutoring use among all 2,000 students. Which is descriptive and which is inferential? Why might another sample give a different proportion?

3. Classify each as categorical, numerical discrete, or numerical continuous. Also give its level of measurement: nominal, ordinal, interval, or ratio. (a) Course subject; (b) number of visits; (c) measured visit length; (d) satisfaction rated poor/fair/good/excellent; (e) room temperature in degrees Celsius.

Sampling and frequency tables

4. Identify the sampling method. (a) Randomly choose 20 names from the full roster. (b) Randomly choose 5 students in each year of study. (c) Randomly choose 3 class sections and survey everyone in those sections. (d) Choose a random starting name and then every 25th name. Which method guarantees representation from every year?

5. A survey link is posted only in the tutoring center and participation is voluntary. Explain two sources of bias if the goal is to estimate college-wide tutoring use. Would collecting 1,000 responses automatically fix them?

6. These 12 students reported numbers of tutoring visits: 0, 1, 2, 0, 3, 1, 1, 2, 0, 4, 2, 0. Build a table of frequency, relative frequency, and cumulative relative frequency for 0 through 4 visits. What fraction reported at most 2 visits?

Evidence, causation, and ethics

7. Students who choose tutoring have higher course averages. Can this observation establish that tutoring caused the difference? Name a plausible confounding variable and explain its role.

8. Design an experiment comparing two problem-practice formats using 60 volunteers. Identify the experimental units, treatments, response variable, and random assignment. Explain why random assignment helps, and why volunteers still limit generalization.

9. A report excludes unfavorable responses and publishes student names next to their ratings without permission. Identify the ethical problems and describe a better reporting practice.